

ANALOG CMOS CIRCUIT DESIGN SS26 - PRACTICAL EXERCISE I

Introduction to Cadence

Tutors

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Group

Name

Matriculation Number

Date

Feedback

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Points **Mark**

Date **Signature**

No review without this standard form!



T1: DC Analysis, DC Sweep, Parameter Sweep

- Calculate the signal transfer function $v_{\text{out}}/v_{\text{in}}$ of the circuit shown in Figure 1.a.
- Calculate V_{out} for $V_{\text{in}} = 0.1\text{V}, 0.2\text{V} \dots 3.3\text{V}$ and $R_1 = R_2 = 1\text{k}\Omega$!
- Calculate V_{out} for $V_{\text{in}} = 3.3\text{V}$, $R_1 = 1\text{k}\Omega$ and $R_2 = 1\text{k}\Omega, 2\text{k}\Omega \dots 10\text{k}\Omega$!

Verify the results of (b) and (c) by means of simulation (dc with parameter sweep) and document the simulation results. Implement resistors by means of “res” from the library “analogLib”, DC voltage sources by means of “vdc” from the library “analogLib” and define ground by means of “gnd” from the library “analogLib”.

T2: Transient Analysis, AC-Analysis

- Calculate the signal transfer function, the magnitude and the phase for $v_{\text{out}}/v_{\text{in}}$ of the circuit shown in Figure 1.b. Verify your results by means of simulation (ac) and document the simulation results for $R = 12.5\text{k}\Omega$ and $C = 8\text{pF}$. Implement capacitors by means of “cap” from the library “analogLib” and transient as well as small signal AC voltage sources by means of “vsin” from the library “analogLib”.
- What is the definition of the -3dB cut-off frequency? Calculate the cut-off frequency for $R = 12.5\text{k}\Omega$ and $C = 8\text{pF}$.
- Sketch $v_{\text{out}}(t)$ and $v_{\text{in}}(t)$ for $v_{\text{in}}(t) = 1\text{V} + 1\text{V} \sin(2\pi \cdot 1.6\text{MHz} \cdot t)$ and $R = 12.5\text{k}\Omega$ and $C = 8\text{pF}$. Explain your approach using the results of (a), then verify your result by means of documented simulation (tran) results.

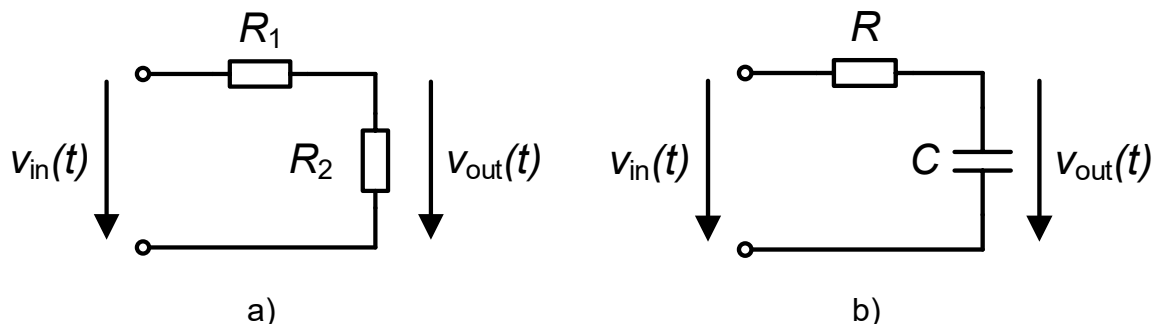


Figure 1: Two simple circuits.

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PRACTICAL EXERCISE I

T3: Transistor Transfer Characteristic Curve

- a) Simulate the I_D versus U_{DS} characteristic curve of an nmos-Transistor ("nmos" from library "PRIMLIB") for $0V < V_{DS} < 3.3V$, $V_{GS} = 0.5V, 0.6V, \dots 1.2V$, a transistor length L equal to $0.35\mu m$ and $1\mu m$ and a transistor width W equal to $2\mu m$. Attach the simulation (parametric sweep) results to the documentation.
- b) Derive an equation which allows for the determination of the channel length modulation parameter λ (lambda) from the simulation results.
- c) Determine λ for the parameter sets of V_{GS} and L given in (a). What can be observed?

